EXECUTIVE REPORT

Comprehensive **Vulnerability** Management Initiative

Qualys-Powered Risk Reduction and Security Posture Enhancement

Presented to C-Level Stakeholders

Ejoke John | Cybersecurity Analyst
ejoke.john4socanalyst@outlook.com

83%	132→22	100%	0
VULNERABILITY REDUCTION	CONFIRMED VULNERABILITIES	CRITICAL ISSUES RESOLVED	DAYS SYSTEM DOWNTIME

EXECUTIVE SUMMARY

The State of Your Security

Your environment was carrying 132 confirmed vulnerabilities, four of them rated Critical, none of which were visible from the outside. A standard external scan of this machine returned just 2 findings and would have given you a false sense of security. Once we ran an authenticated scan with full internal access, the real picture emerged: outdated software, unpatched operating systems, legacy protocols still active, and security policy gaps that an attacker could exploit without sophisticated tools. This report walks you through what was found, what was fixed, and what the numbers look like now.

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Why This Matters to Your Organisation

Unpatched vulnerabilities are consistently among the top entry points attackers exploit. Two of the four Critical findings on your system were active remote code execution vulnerabilities, meaning an attacker with network access could have taken full control of that machine without any user clicking anything. The IBM Security Cost of a Data Breach Report 2024 puts the average breach cost at $4.45 million. The remediation work covered in this report addressed those risks directly, at a fraction of that cost, with zero operational downtime.

Scope and Approach

- Deployed Qualys VMDR with a credentialed scanner to achieve full internal visibility across your Windows 10 environment
- Ran a baseline unauthenticated scan to establish what an external attacker would see, then followed with a full authenticated scan to uncover the complete risk profile
- Prioritised remediation by severity, starting with Critical and High findings, to achieve maximum risk reduction per effort
- Ran a verified post-remediation scan to confirm the improvements with evidence, not assumption
- Translated every finding into business risk language so your leadership team can make informed decisions without reading raw scan output

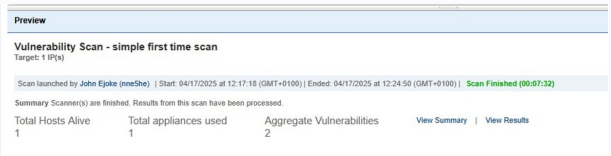
ASSESSMENT ENVIRONMENT

What We Scanned and How

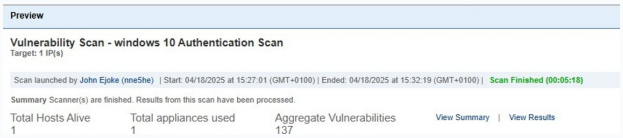
Component	Details
Scanning Platform	Qualys Community Edition (qualysguard.qg2.apps.qualys.eu)
Scanner	Qualys Virtual Scanner Appliance in VirtualBox (JohnsVA)
Target	Windows 10 Home 64-bit, IP 172.20.10.2 (DESKTOP-O8C0H2E)
Network Configuration	Oracle VirtualBox, Bridged Adapter, scanner and target on same subnet
Known Risk Conditions	VLC 1.1.1, Firefox 94.0, SMBv1 enabled, weak security policy in place

External View vs Internal Reality

The single most important finding in this engagement is the gap between what your environment looks like from the outside and what it actually contains. An unauthenticated scan, the type most organisations rely on, found 2 vulnerabilities. The authenticated scan with internal credentials found 132. That gap represents 130 confirmed vulnerabilities your perimeter scan would never surface. Below is the direct comparison.



Unauthenticated scan: 2 findings visible from the network



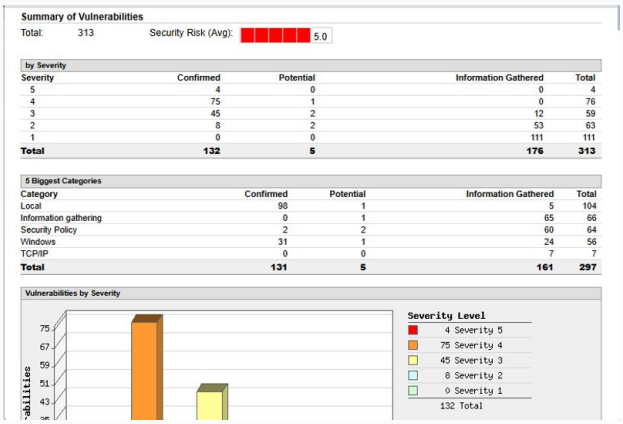
Authenticated scan: 137 findings with full internal access

Metric	Unauthenticated	Authenticated
Confirmed Vulnerabilities	2	132
Severity 5 (Critical)	0	4
Severity 4 (High)	0	75
Severity 3 (Medium)	1	45
Security Risk Average	3.0	5.0

> *If your security programme relies solely on external scanning, you are not seeing the real risk profile of your environment. Authenticated scanning is not optional. It is the baseline for accurate visibility.*

PRE-REMEDIATION FINDINGS

What Your Authenticated Scan Revealed



Summary of Vulnerabilities

Total: 313 Security Risk (Avg): 5.0

by Severity

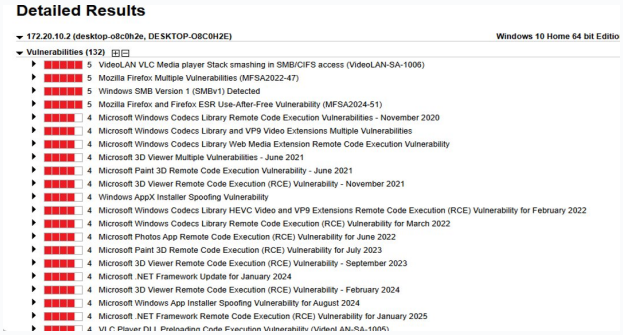
Severity	Confirmed	Potential	Information Gathered	Total
5	4	0	0	4
4	75	1	0	76
3	45	2	12	59
2	8	2	53	63
1	0	0	111	111
Total	**132**	**5**	**176**	**313**

5 Biggest Categories

Category	Confirmed	Potential	Information Gathered	Total
Local	98	1	5	104
Information gathering	0	1	65	66
Security Policy	2	2	60	64
Windows	31	1	24	56
TCP/IP	0	0	7	7
Total	**131**	**5**	**161**	**297**

Vulnerabilities by Severity

Scan summary: 313 total findings, 132 confirmed, risk average 5.0



Detailed Results

172.20.10.2 (desktop-o8c0h2e, DESKTOP-O8C0H2E) Windows 10 Home 64 bit Editio

Vulnerabilities (132)

- 5 VideoLAN VLC Media player Stack smashing in SMB/CIFS access (VideoLAN-SA-1006)
- 5 Mozilla Firefox Multiple Vulnerabilities (MFSA2022-47)
- 5 Windows SMB Version 1 (SMBv1) Detected
- 5 Mozilla Firefox and Firefox ESR Use-After-Free Vulnerability (MFSA2024-51)
- 4 Microsoft Windows Codecs Library Remote Code Execution Vulnerabilities - November 2020
- 4 Microsoft Windows Codecs Library and VP9 Video Extensions Multiple Vulnerabilities
- 4 Microsoft Windows Codecs Library Web Media Extension Remote Code Execution Vulnerability
- 4 Microsoft 3D Viewer Multiple Vulnerabilities - June 2021
- 4 Microsoft Paint 3D Remote Code Execution Vulnerability - June 2021
- 4 Microsoft 3D Viewer Remote Code Execution (RCE) Vulnerability - November 2021
- 4 Windows AppX Installer Spoofing Vulnerability
- 4 Microsoft Windows Codecs Library HEVC Video and VP9 Extensions Remote Code Execution (RCE) Vulnerability for February 2022
- 4 Microsoft Windows Codecs Library Remote Code Execution (RCE) Vulnerability for March 2022
- 4 Microsoft Photos App Remote Code Execution (RCE) Vulnerability for June 2022
- 4 Microsoft Paint 3D Remote Code Execution (RCE) Vulnerability for July 2023
- 4 Microsoft 3D Viewer Remote Code Execution (RCE) Vulnerability - September 2023
- 4 Microsoft .NET Framework Update for January 2024
- 4 Microsoft 3D Viewer Remote Code Execution (RCE) Vulnerability - February 2024
- 4 Microsoft Windows App Installer Spoofing Vulnerability for August 2024
- 4 Microsoft .NET Framework Remote Code Execution (RCE) Vulnerability for January 2025

Detailed findings: 4 Critical, 75 High, 45 Medium severity confirmed

CRITICAL AND HIGH SEVERITY FINDINGS REQUIRING IMMEDIATE ACTION

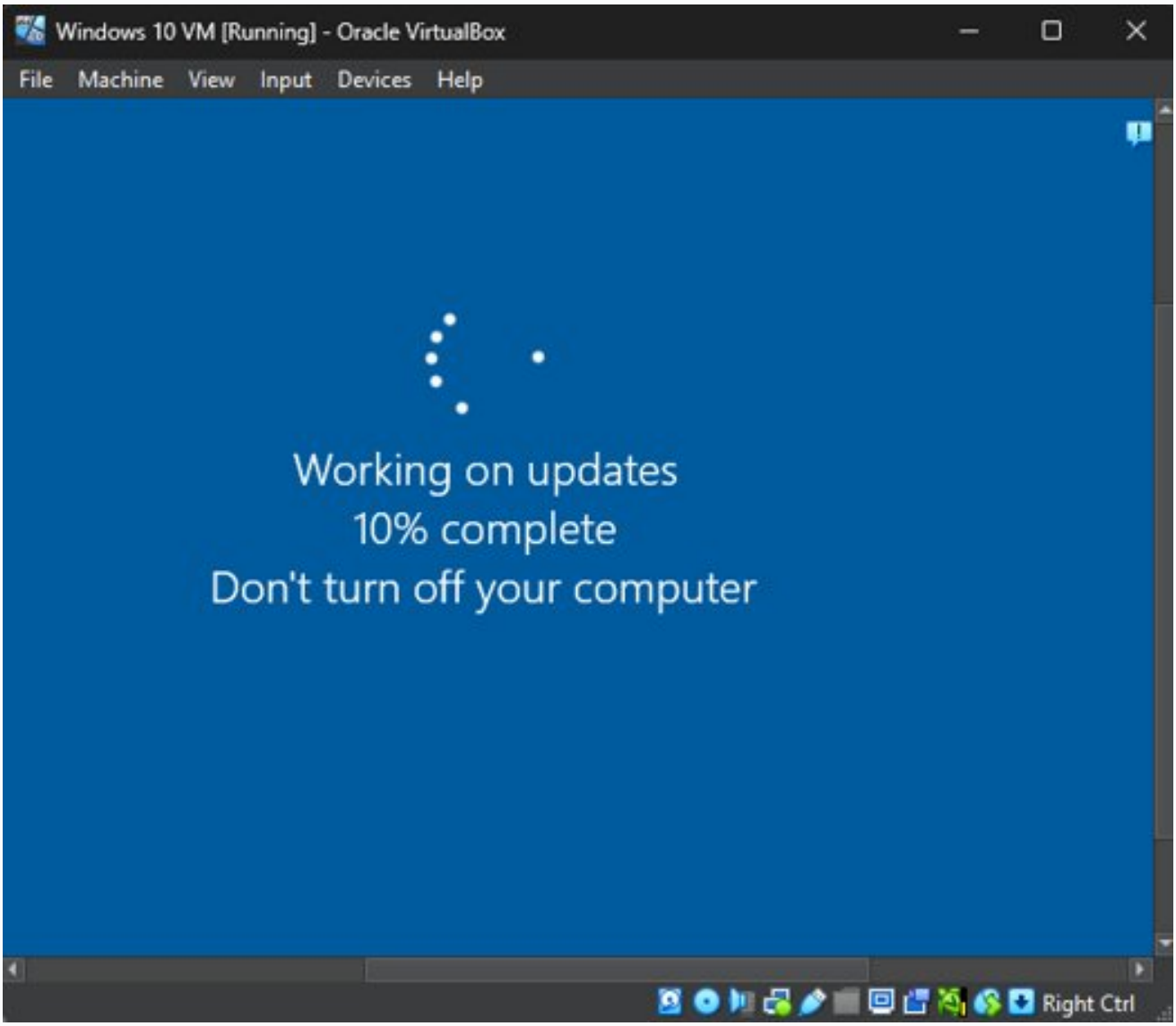
Sev	Finding	Business Risk
5	VLC 1.1.1 Stack Smashing (VideoLAN-SA-1006)	Remote code execution: attacker takes full control via crafted file
5	Firefox Multiple Vulnerabilities (MFSA2022-47)	Remote code execution and privilege escalation from browser
5	Firefox Use-After-Free (MFSA2024-51)	Memory corruption leading to arbitrary code execution
5	Windows SMBv1 Protocol Active	Full system compromise via EternalBlue-class exploit, no user interaction needed
4	MS Windows Codecs Library RCE, Nov 2020	Remote code execution via crafted media file
4	MS 3D Viewer Vulnerabilities, Jun 2021	Remote code execution
4	Windows AppX Installer Spoofing	Privilege escalation via malicious installer
4	MS .NET Framework RCE, Jan 2025	Code execution in .NET runtime
4	VLC DLL Preloading RCE (VideoLAN-SA-1005)	Code execution via malicious DLL sideloading
3	SMBv2 Message Signing Not Required	Credential theft via SMB relay attack
2	NetBIOS Name Service Accessible	Network reconnaissance and name poisoning
2	Cached Logon Credentials Enabled	Offline credential extraction if disk is accessed

REMEDIATION AND VERIFICATION

What Was Done and What Changed

Remediation was executed in strict severity order. Every Critical finding on your system traced back to two applications that had not been updated: VLC 1.1.1 and Firefox 94.0. Replacing both with current versions removed all four Severity 5 vulnerabilities in a single step. Windows Update was then applied to address the 75 High severity Microsoft CVEs. Protocol and policy hardening closed the remaining network-level exposure.

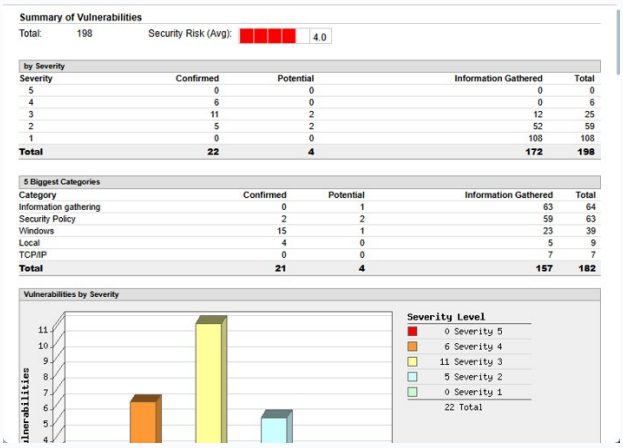
Action Taken	What Was Done	Risk Closed
Software Upgrade	Removed VLC 1.1.1 and Firefox 94.0, installed current secure versions	All 4 Critical findings
OS Patching	Applied all outstanding Windows security updates via Windows Update	75+ High severity CVEs
Protocol Hardening	Disabled SMBv1, enforced SMB message signing, disabled NetBIOS over TCP/IP	SMBv1, SMBv2 signing, NetBIOS
Policy Hardening	Strengthened password policy, disabled legacy Telnet, enforced access controls	Policy-level findings



Windows Update applying critical security patches during the remediation phase

Verification Scan Results

A full authenticated scan was run after every remediation action was complete. This is the evidence behind the numbers, not an estimate or a checklist, but a live scan of your environment showing exactly what remains.



Summary of Vulnerabilities

Total: 198 Security Risk (Avg): 4.0

by Severity

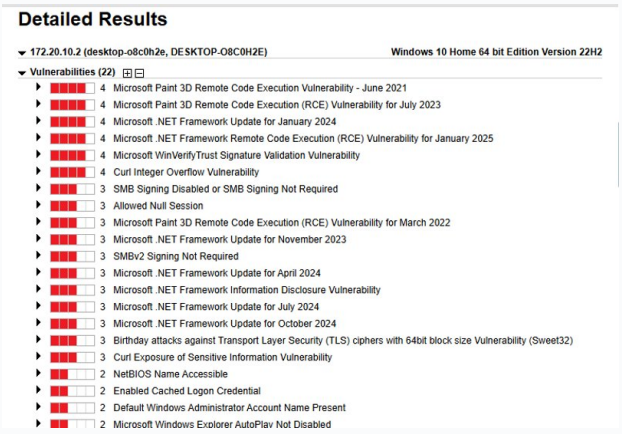
Severity	Confirmed	Potential	Information Gathered	Total
5	0	0	0	0
4	6	0	0	6
3	11	2	12	25
2	5	2	52	59
1	0	0	108	108
Total	**22**	**4**	**172**	**198**

5 Biggest Categories

Category	Confirmed	Potential	Information Gathered	Total
Information gathering	0	1	63	64
Security Policy	2	2	59	63
Windows	15	1	23	39
Local	4	0	5	9
TCP/IP	0	0	7	7
Total	**21**	**4**	**157**	**182**

Vulnerabilities by Severity

After remediation: 22 confirmed vulnerabilities, zero Severity 5 findings



Detailed Results

172.20.10.2 (desktop-o8c0h2e, DESKTOP-O8C0H2E) Windows 10 Home 64 bit Edition Version 22H2

Vulnerabilities (22)

- 4 Microsoft Paint 3D Remote Code Execution Vulnerability - June 2021
- 4 Microsoft Paint 3D Remote Code Execution (RCE) Vulnerability for July 2023
- 4 Microsoft .NET Framework Update for January 2024
- 4 Microsoft .NET Framework Remote Code Execution (RCE) Vulnerability for January 2025
- 4 Microsoft WinVerifyTrust Signature Validation Vulnerability
- 4 Curl Integer Overflow Vulnerability
- 3 SMB Signing Disabled or SMB Signing Not Required
- 3 Allowed Null Session
- 3 Microsoft Paint 3D Remote Code Execution (RCE) Vulnerability for March 2022
- 3 Microsoft .NET Framework Update for November 2023
- 3 SMBv2 Signing Not Required
- 3 Microsoft .NET Framework Update for April 2024
- 3 Microsoft .NET Framework Information Disclosure Vulnerability
- 3 Microsoft .NET Framework Update for July 2024
- 3 Microsoft .NET Framework Update for October 2024
- 3 Birthday attacks against Transport Layer Security (TLS) ciphers with 64bit block size Vulnerability (Sweet32)
- 3 Curl Exposure of Sensitive Information Vulnerability
- 2 NetBIOS Name Accessible
- 2 Enabled Cached Logon Credential
- 2 Default Windows Administrator Account Name Present
- 2 Microsoft Windows Explorer AutoPlay Not Disabled

Remaining findings: highest severity is now 4, no critical exposure

RESULTS AND BUSINESS VALUE

Your Risk Profile: Before and After

Metric	Before	After	Change
Confirmed Vulnerabilities	132	22	**-83%**
Critical (Severity 5)	4	0	**-100%**
High (Severity 4)	75	6	**-92%**
Medium (Severity 3)	45	11	**-76%**
Security Risk Average	5.0	4.0	**-1.0**
Total Aggregate Findings	313	198	**-37%**

What These Numbers Mean for Your Business

Every Critical vulnerability on your system was an active, exploitable risk. Not theoretical. Not edge-case.. Two of the four could have been triggered remotely with no user interaction. Closing them directly reduces your probability of the kind of incident that costs organisations an average of $4.45 million in recovery, legal fees, and regulatory penalties (IBM Security, 2024). The work done here is a foundation, not a finish line. Your organisation's exposure is materially lower today than it was at the start of this engagement.

Business Benefit	What It Means for Your Organisation
Breach Cost Reduction	Four Critical RCE vectors closed. The highest-probability breach scenarios on your system are now resolved
Operational Continuity	Reduced attack surface directly lowers ransomware risk, which is responsible for the majority of operational downtime incidents
Compliance Readiness	Remediated findings align your posture with GDPR, HIPAA, and PCI DSS vulnerability management requirements
Efficient Security Investment	Severity-first prioritisation means your team's effort is always directed at the highest-impact risks first
Auditable Evidence	Before and after scan data provides documented proof of remediation, usable in audits, board reporting, and insurance assessments

STRATEGIC RECOMMENDATIONS

Where Your Programme Needs to Go Next

Closing these vulnerabilities was the first step. Sustaining this posture requires a continuous programme, not periodic clean-up. Below are five recommendations for building that programme, ordered by priority.

01 MOVE TO CONTINUOUS SCANNING

Your current posture was assessed at a point in time. New vulnerabilities are disclosed daily. Automating continuous scanning within Qualys VMDR means your risk picture is always current, not six months out of date.

02 CONNECT SCANNING TO PATCH MANAGEMENT

Today, a vulnerability is found and a human decides to act. Integrating Qualys with your patch management system removes that gap. When a Critical finding appears, patching begins automatically, cutting your window of exposure from weeks to hours.

03 EXPAND CREDENTIALED SCANNING ACROSS ALL ASSETS

This engagement covered one machine. Your production estate, including servers, cloud instances, and endpoints, carries the same risk of hidden vulnerabilities. Authenticated scanning must cover all critical assets, not just a subset.

04 BUILD EXECUTIVE-LEVEL RISK DASHBOARDS

Leadership should not need to read a scan report to understand your security posture. A live dashboard tracking time-to-remediate, critical finding count, and compliance adherence gives you the visibility to make decisions and answer board questions in real time.

05 IMPLEMENT SCHEDULED CONFIGURATION AUDITING

Vulnerabilities are not the only risk. Configuration drift, such as a service re-enabled, a policy weakened, or a default credential left in place, creates exploitable gaps that a vulnerability scan alone will not catch. Quarterly CIS benchmark audits close that gap.

IMPLEMENTATION TIMELINE

Horizon	Recommended Actions
30 days	Deploy continuous authenticated scanning across all production assets
90 days	Integrate patch management automation, expand credentialed scan coverage
6 months	Launch executive risk dashboards and implement quarterly configuration auditing

CONCLUSION

Where Your Organisation Stands

Before this engagement, your environment carried 132 confirmed vulnerabilities, four of them Critical, none of them visible from a standard external scan. Your attack surface was significantly larger than your existing security tools were showing you. That gap between perceived risk and actual risk is one of the most dangerous positions an organisation can be in, because it creates confidence in a security posture that does not exist.

The remediation work completed here closed 83% of those confirmed vulnerabilities, eliminated every Critical finding, and produced verified, auditable evidence of the improvement. Your security risk average dropped a full point. Your environment is measurably safer today than it was at the start of this engagement, with zero downtime and no disruption to operations.

The work is not finished. Twenty-two confirmed vulnerabilities remain, and your broader estate has not yet been assessed to this depth. The recommendations in this report are the roadmap for closing the remaining gap and building a programme that keeps your risk profile current, not just periodically improved.

RISK REDUCTION
110 of 132 confirmed vulnerabilities eliminated, attack surface reduced by 83%

CRITICAL EXPOSURE CLOSED
All four Severity 5 findings resolved, zero critical risk remaining in the verified scan

BREACH COST AVOIDANCE
Highest-probability exploit vectors closed, directly reducing likelihood of a $4.45M breach event

COMPLIANCE ALIGNMENT
Remediated posture supports GDPR, HIPAA, and PCI DSS vulnerability management obligations

OPERATIONAL CONTINUITY
Full remediation cycle completed with zero operational disruption or downtime

AUDITABLE EVIDENCE
Before and after scan data documented, available for audit, board reporting, and insurance review

Ejoke John
Cybersecurity Analyst

ejoke.john4socanalyst@outlook.com
linkedin.com/in/john-ejoke
github.com/john-ejoke